CLIFF

CLIFF-LMD
CLIFF-RMD
CLIFF_LED_EN
CLIFF-RRD
CLIFF-RFD
CLIFF-LRD
CLIFF-LFD

CLIFF-LM
CLIFF-RM

File: CLIFF.kicad_sch

BUMPER

BUMP-L
BUMP-R
BUMP_L_LED_EN
BUMP_R_LED_EN

File: bumper_sensors.kicad_sch

SIDE SENSORS

VSENS-3V3
SIDE-R1
ZX-SH1_0-7PWT

USART2-IR-R-RX
SIDE-PROXI-CE-R
I2C4_SCL
I2C4_SDA

CARPET SENSOR

LO
ECHO_ADCD

CARPET-SENSOR-LOD
CARPET-SENSOR-ECHO

File: Carpet-sensor.kicad_sch

MAIN BRUSH MOTOR

MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-ADCD

File: MAIN-BRUSH-MOTOR .kicad_sch

WATCHDOG

WDI
WD_OKD

File: WATCHDOG.kicad_sch

WHEEL MOTORs

WHEEL-MOT-R-IN2
WHEEL-MOT-R-IN1
WHEEL-MOT-R-ENCODED
WHEEL-MOT-R-DROPD
WHEEL-MOT-L-IN2
WHEEL-MOT-L-IN1
WHEEL-MOT-L-DROPD
WHEEL-MOT-L-ENCODED
WHEEL-MOT-R-ADCD
WHEEL-MOT-L-ADCD

File: WHEEL-MOTORs .kicad_sch

LiDAR

WD_OK
LiDAR-RXD
LiDAR-TXD
LIDAR-MOTOR-CTRL
LiDAR-EN

LIDAR_RXD
LIDAR_TX
LIDAR-MOTOR-CTRLD

File: LIDAR.kicad_sch

MAIN FAN

MAIN-FAN-FBD
MAIN-FAN-PWM

File: MAIN-FAN .kicad_sch

USB-C POWER

DP
DN
USB_CC1
USB_CC2
USB_VBUS_SENSE
USB_PWR_EN

USB_CC1D
USB_CC2
USB_VBUS_SENSE
USB_PWR_END

File: usb_power.kicad_sch

SIDE BRUSH MOTORs

SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-ADCD

File: SIDE-BRUSH-MOTORs .kicad_sch

MCU-STM32

VMOT-14V4-EN
VL-I2C-RESETD
UART3_RX2
VL-L-LPnD
UART3_RX1

File: STM32G473.kicad_sch

FRONT SENSORS

VSENS-3V3
VL-R1
2005FV-2X4P
R196
47k
VL-R-LPn
VL-I2C-RESET
IR-F-R
VL-L1
VL-L-LPn
IR-F-L
C6
10uF/10V
C19
10uF/10V

IMU

INT1D
SDA
SCL

File: IMU-ICM-4267-P .kicad_sch

POWER CONTROL

VMOT-14V4-EN
STM-PWR-CTRL
PWR-BTN-SENSE
VSENS_3V3_EN
VSENS-5V-USB-EN
WD_OK
PWR-SW-G
POWER-BTN

File: SYSTEM-POWER.kicad_sch

BUTTON-LEDs

POWER-BTND
LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND

File: BUTTON-LEDs .kicad_sch

BATTERY CHARGER

BATTERY-CHARGER
PWR-SW-G
BAT_SENSE
DC_IN_SENSE
ID

File: Battery-Charger.kicad_sch

HALL-BIN1
HALL-TOWER1
ZX-SH1_0-3PLT
HALL-BIN
HALL-TOWER

REAR-IR1
ZX-SH1_0-4PLT
REAR-IR-L
REAR-IR-R

R2
47k
R6
2.2k
R5
2.2k

INT-SENSORS
WDID

Sheet: /
File: Main.kicad_sch

Title:

Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/16

5V USB, SENSORS, LiDAR POWER RAIL
– set 5V_EN HIGH to enable

TPS54560B (U21, U22):
– non–synchronous
– COMP 4.7k + 47nF (all–ceramic Cout, no pole cap)
– SS56 catch diode; Fsw 418 kHz (RT 232k)

R146 10
U22 TPS54560BDDAR
C119 100nF/50V
VBAT–14V4
PWR_FLAG
VSENS–5V–USB–END
C115 10uF/50V
C116 10uF/50V
C117 100nF/50V
R206 232k
R168 100k
PWR_FLAG
L3 3.3uH
D43 SS56
R148 12k
R149 2.2k
R147 4.7k
C120 47nF/50V
C121 10uF/10V
C122 10uF/10V
C147 10uF/10V
VSENS–5V–USB
LED1 YLED0603B
R89 1k

TPS54560B RT: fsw = 92417 x RT(k)^–0.991 = 418 kHz

VBAT–14V4
Q1 AO4407C
PWR_FLAG
VMOT–14V4
C62 10uF/25V
D37 SS34
VBAT–14V4
Q37 AO4407C
VMOT–14V4
10k R165
R36 10
R222 10
VMOT–SW–G
VMCU–3V3
10k R184
Q19 AO3400
WD_OK
VMOT–14V4–END

MOTOR POWER RAIL 14.4V

– on/off motor power, no soft ramp
– set VMOT–14V4–EN LOW to enable power
– floating (MCU flash/reset) disables power
– WD_OK LOW shuts power off

VMOT–14V4: Q1 + Q37 AO4407C in parallel
– ~7.5 A normal (with the MSD–G fan) is ~0.25 W each
– short stall peaks (~13 A) fine with copper under the drain
– Vgs +/–25 V, no gate divider needed

C176 gate–source cap, 10x C64: holds Q16 off while VCC–3V3 ramps up

VMCU–3V3
Q16 AO3401
PWR_FLAG
VSENS–3V3
100k R90
C176 1uF/25V
LED3 YLED0603B
R98 1k
R195 10k
C64 100nF/50V
VSENS_3V3_EN

3.3V SENSORS POWER RAIL

– C176 keeps Q16 off when power switches ON
– C64 soft–ramps power
– set VSENS_3V3_EN LOW to enable sensors power

VBAT–14V4
Q13 AO4407C
VMCU–3V3–POWER–EN
R16 100k
R67 10k
C14 1uF/25V
R171 100k
PWR–SW–G
POWER–BTN
C22 1uF/25V
D26 1N4148
VMCU–3V3
10k R173
PWR–BTN–SENSE
VMCU–3V3
10k R172
Q23 AO3400
STM–PWR–CTRLD
R190 1k
C21 10uF/10V
PWR–LATCH
VMCU–3V3
R193 1M
R191 100k
Q29 AO3400
R192 47k
C172 10uF/10V
VMCU–3V3
C173 100nF/50V
U23 74LVC1G17
Q30 AO3400

Power off –> user pushes POWER–BTN –> Q13 enables MCU 3.3V rail –> MCU immediately drives STM–PWR–CTRL HIGH to keep MCU 3.3V rail power on –> MCU enables other power rails as needed
Power on –> user pushes POWER–BTN –> MCU detects PWR–BTN–SENSE press –> firmware acts as needed, including turning off all power rails including MCU 3.3V itself by driving STM–PWR–CTRL low
Power on –> MCU hung –> user pushes POWER–BTN for ~5 seconds –> Q13 shuts off MCU 3.3V rail –> all other power rails shut off automatically when MCU GPIOs float

Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR–LATCH low). Firmware turns off at 3 s.
While charging, the rail–hold (Q31 on BATTERY–CHARGER, from DC–IN–20V) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR–SW–G low).

3.3V MCU POWER RAIL

– POWER_EN is 14.4V
– set POWER_EN HIGH to enable

TPS54560B (U21, U22):
– non–synchronous
– COMP 4.7k + 47nF (all–ceramic Cout, no pole cap)
– SS56 catch diode; fsw 418 kHz (RT 232k)
– EN abs max 8.4 V: R207/R208 on U21

R141 10
U21 TPS54560BDDAR
VBAT–14V4
C111 100nF/50V
R207 100k
R208 22.1k
C106 10uF/50V
C95 10uF/50V
C102 100nF/50V
R205 232k
L2 3.3uH
D42 SS56
R144 69.8k
R143 4.7k
C112 47nF/50V
R145 22.1k
PWR_FLAG
VMCU–3V3
C113 10uF/10V
C114 10uF/10V
C146 10uF/10V
LED2 YLED0603B
R14 1k

14.5V ULTRASONIC CARPET SENSOR POWER

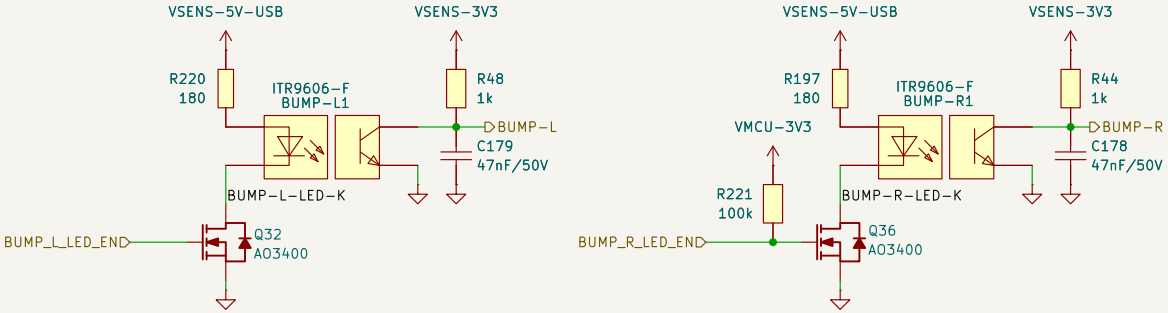
L4 10uH
D4 SS34
PWR_FLAG
VUSONIC–14V4
VSENS–5V–USB
U5
NC SW GND IN FB EN
MT3608
R160 232k
10k R159
C149 10uF/10V
C148 10uF/10V
C150 10uF/25V
C151 10uF/25V

VUSONIC–14V4 = 0.6V x (1 + 232k/10k) ~= 14.5V

VBAT–14V4
MCU VBAT
– Low quiescent current 2.5µA
– Zener diode shuts off when main battery is low
ZMM9V1_C2891750
D2
PWR_FLAG
U16 HT7533–1_C14289
Vout Vin GND
VMCU–BAT–3V3
C140 10uF/25V
C141 10uF/25V

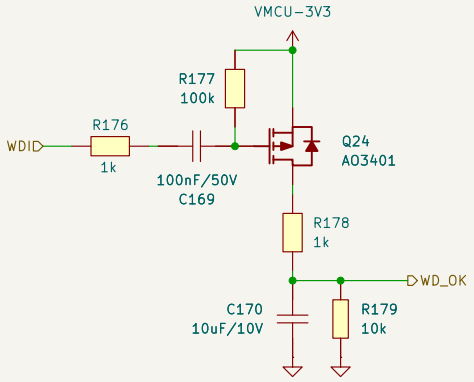
MP1 MP2 MP3 MP4 MP5 MP6

Sheet: /POWER/
File: SYSTEM–POWER.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 2/16



- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range
  1k keeps photo transistor output out of saturation
  so PA4/PA5 (ADC2) see the transition (ITR9606 IC(ON) 0.5-10 mA)
- R221 enables MCU USB-C Rd on both CC1 and CC2 during MCU *reset*,
  so the phone detects the robot whichever way the cable is plugged in

Sheet: /Bumper Sensors/ File: bumper_sensors.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/16



RC CHARGE-PUMP WATCHDOG

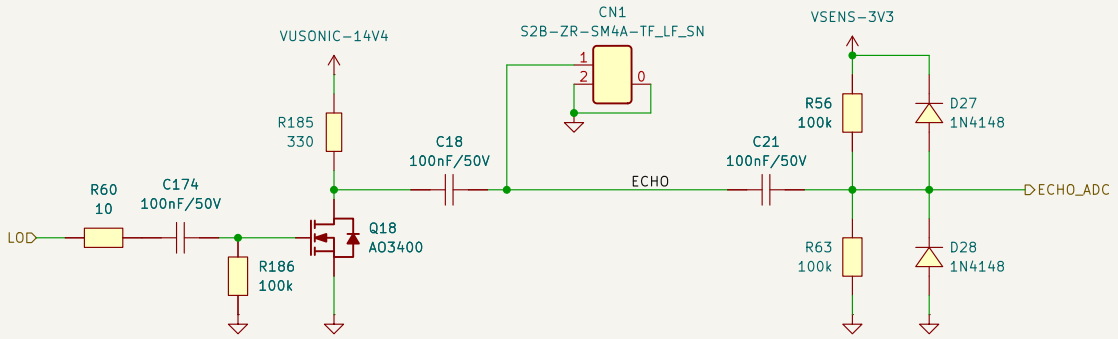
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/16

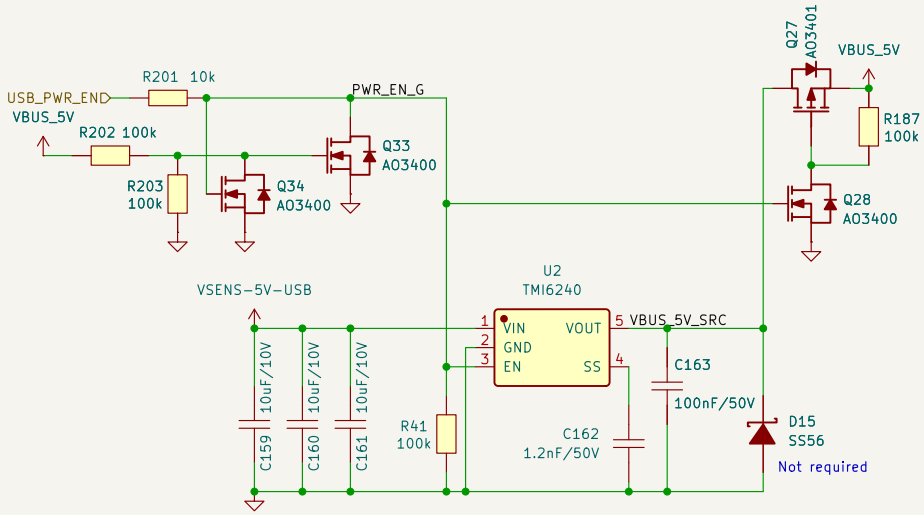
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



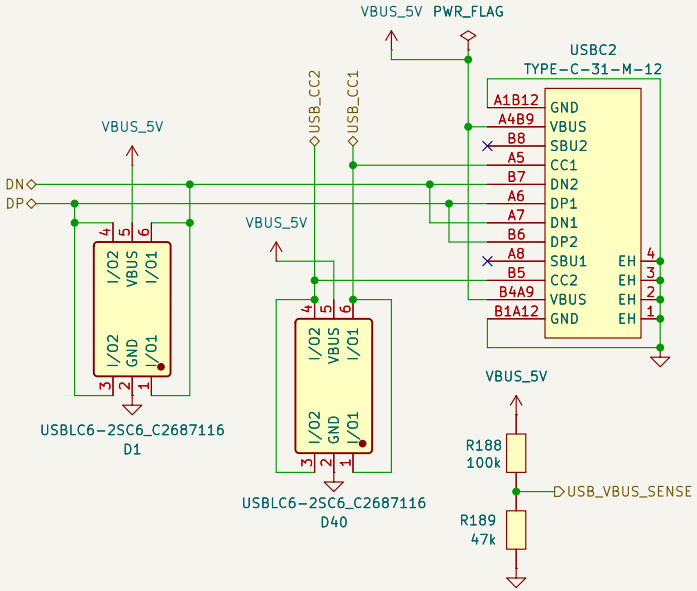
Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/16

USB-C for SMARTPHONE



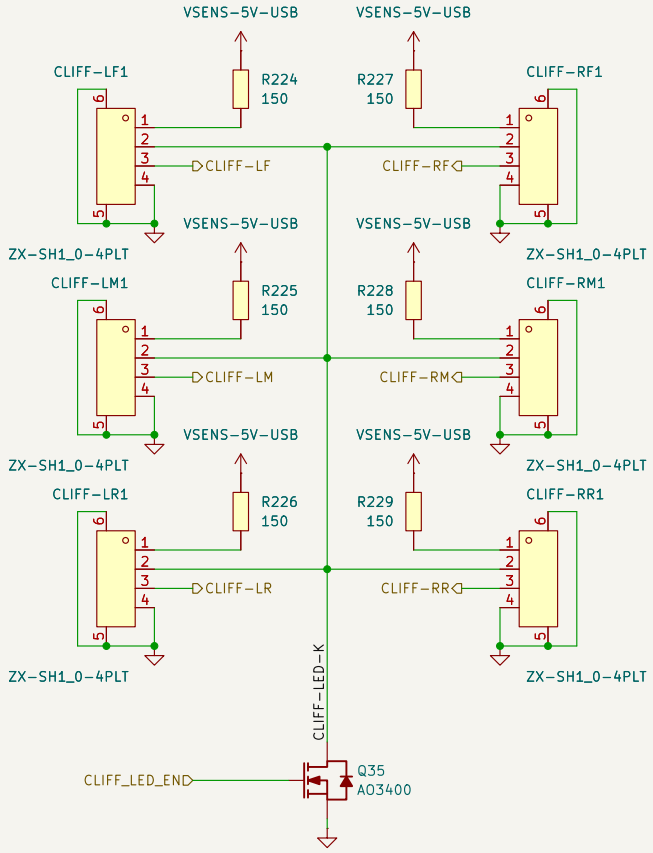
VBUS_5V lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS_5V is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS_5V; once on, Q34 latches it.

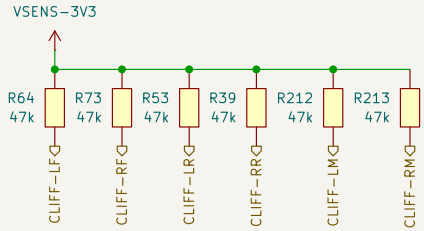
PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS_5V from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); USB_PWR_EN turns on TMI6240 and Q27 together.
TMI6240 now runs from VSENS-5V-USB (U25 buck removed).



USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default.

Sheet: /USB Power/	
File: usb_power.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 7/16



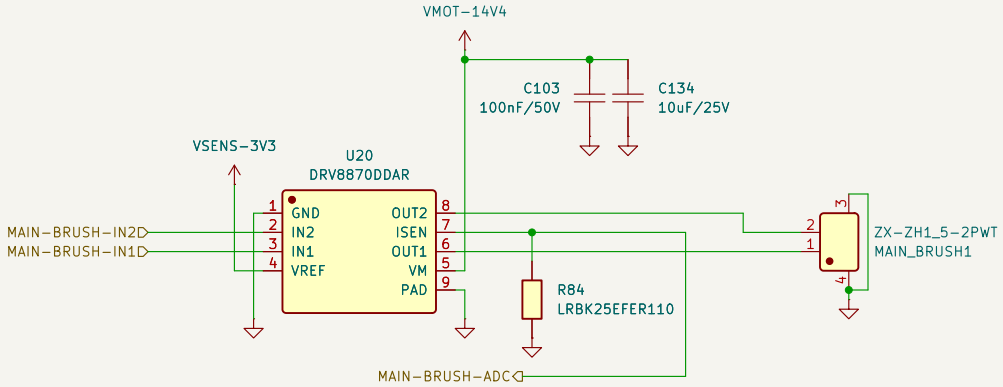


- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light
- possible TODO - cliff LEDs 2x or 3x in series, save resistors
  - worsens power rail noise rejection
  - ADC fw could reject noise by correlating the 6 readings
- 0.76W power consumption when enabled continuously
  - drive at ~10% duty cycle to save power
- tweak output pullup
  - a low resistor value can saturating the photo transistor
  - 47K pullup makes for 1ms output settling time
  - can use 10k for faster settling, to reduce duty cycle further
- combine low duty with fast duty LED cycling
  - fast (~10x photo transistor settling time) reduces LED average radiance to avoid photo transistor saturation
- check sensor faults - is output 3.3V or floating?
  - normal no-cliff condition - photo transisstor gets LED light
  Vary LED current and check photo-transistor voltage.

Sheet: /CLIFF/
File: CLIFF.kicad_sch

Title:

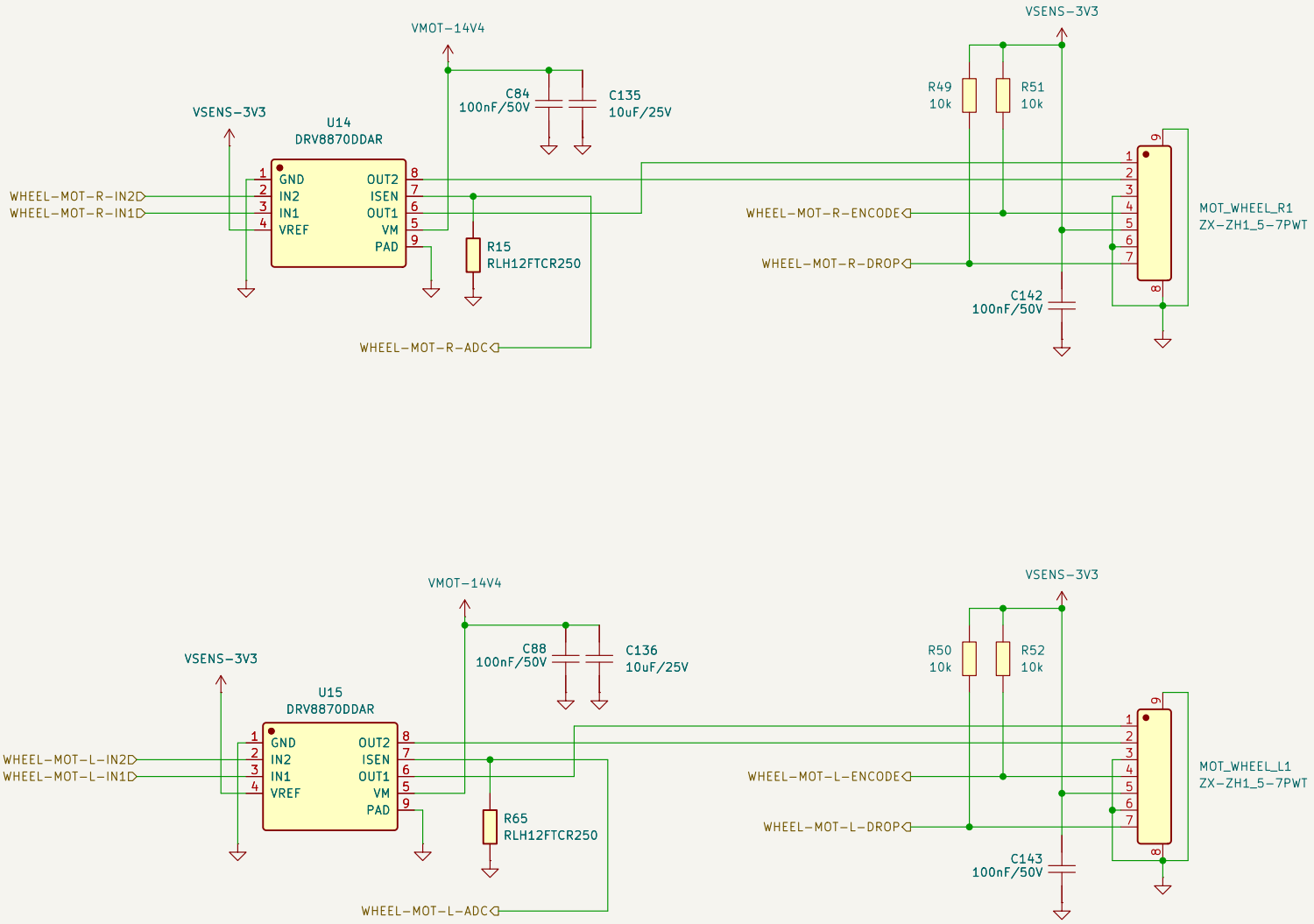
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 19/16



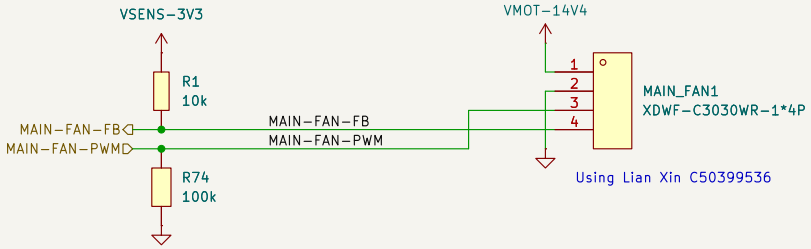
Sheet: /MAIN-BRUSH MOTOR/
File: MAIN-BRUSH-MOTOR .kicad_sch

Title:

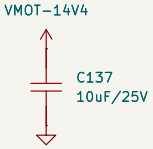
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 22/16



Sheet: /WHEEL MOTOR/	
File: WHEEL-MOTORs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 23/16



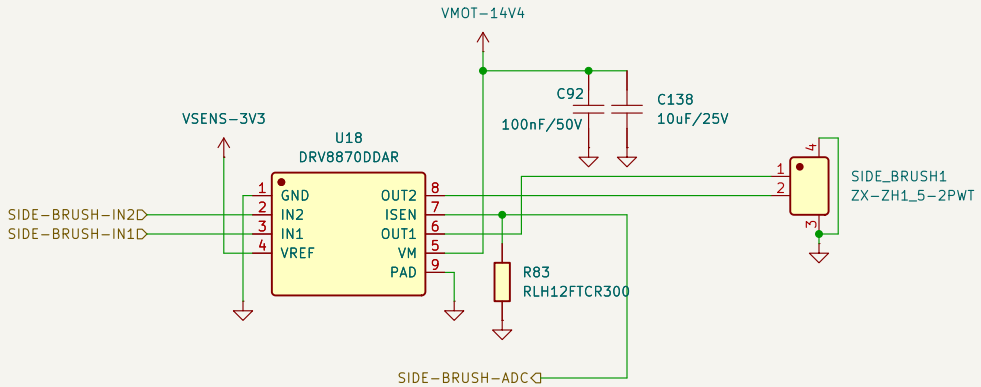
- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB



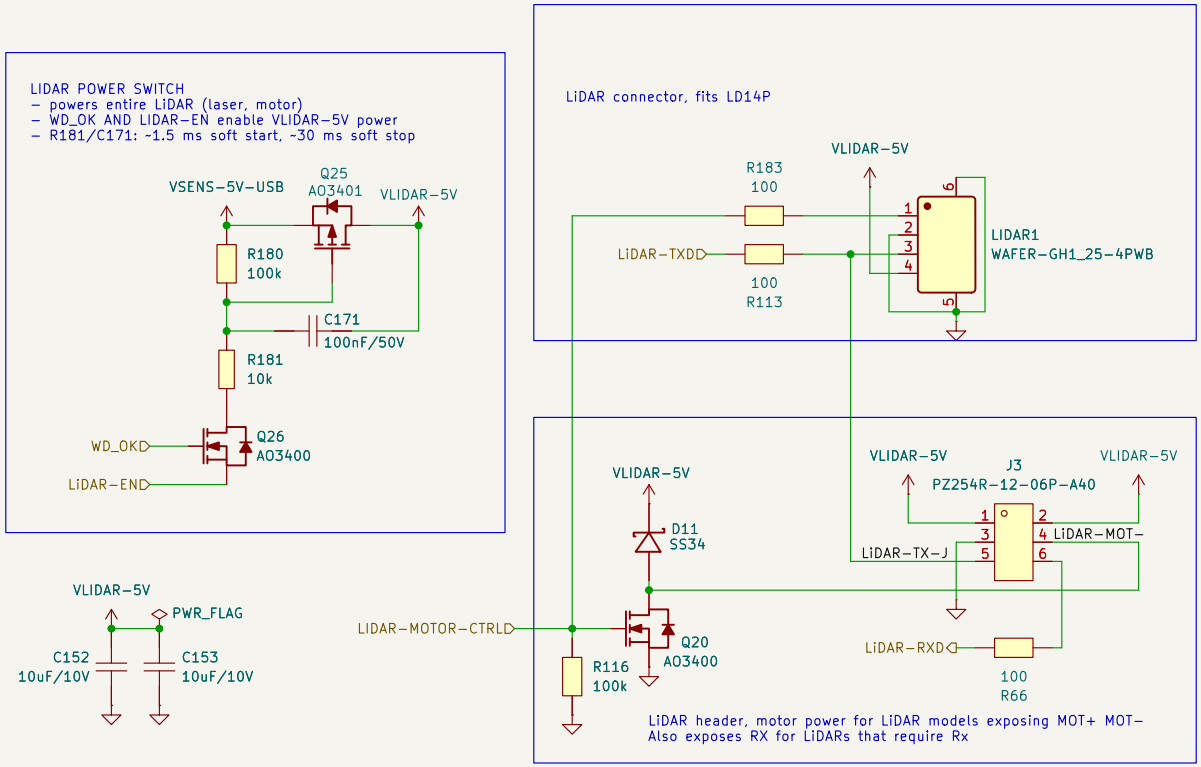
Sheet: /MAIN-FAN/
File: MAIN-FAN .kicad_sch

Title:

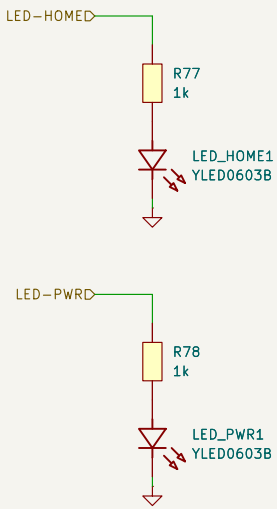
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/16

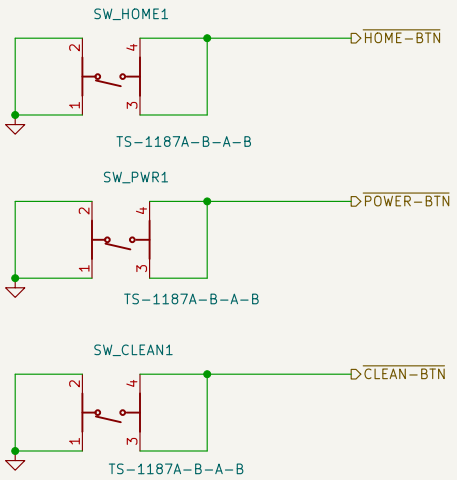


Sheet: /SIDE-BRUSH/	
File: SIDE-BRUSH-MOTORs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 25/16

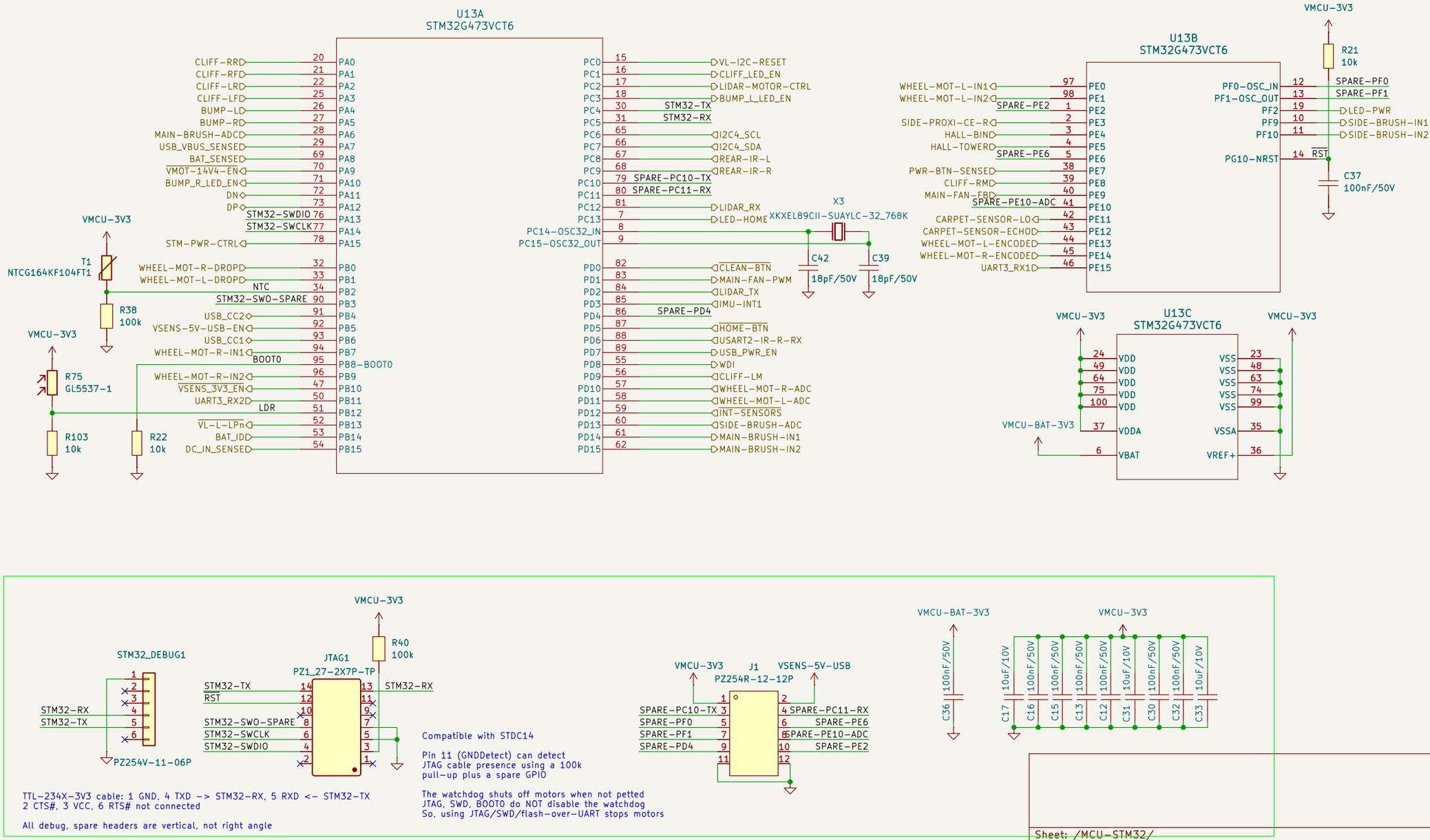


Sheet: /LIDAR/	
File: LIDAR.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 26/16





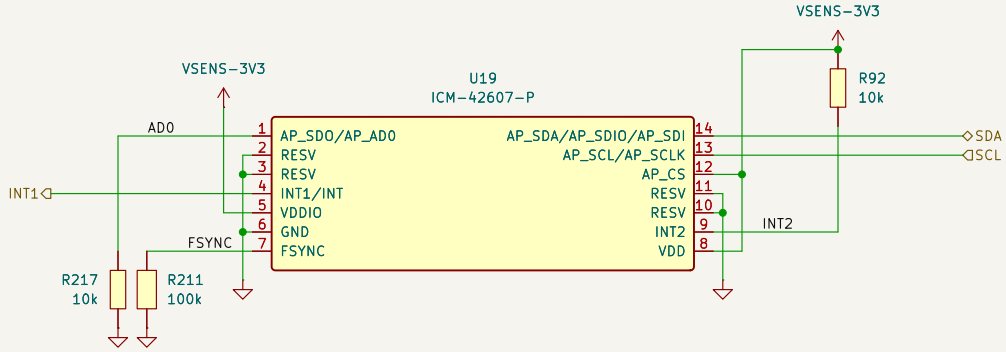
Sheet: /BUTTON-LED/	
File: BUTTON-LEDs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 28/16



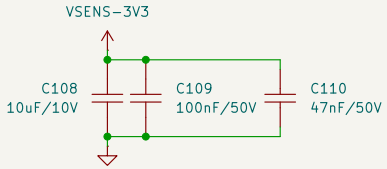
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch

Title:

Size: USLegal	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 30/16



IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

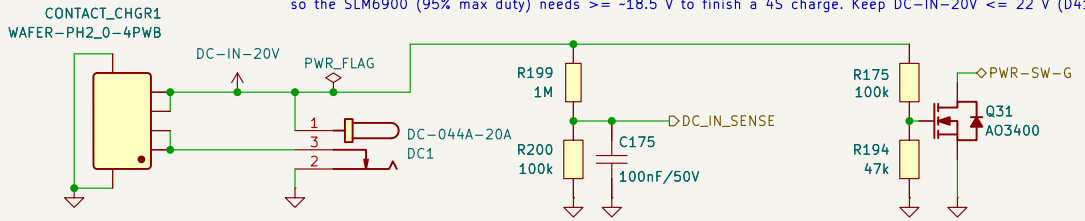


Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

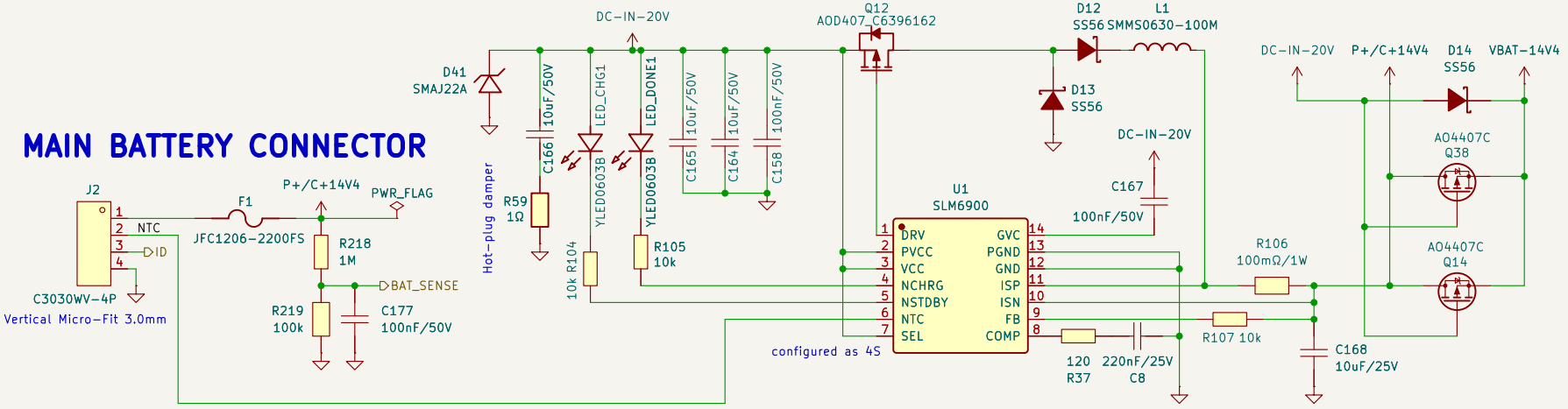
Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 32/16

DC-IN-20V: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT-14V4 = DC-IN-20V - 0.4 V,
so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC-IN-20V <= 22 V (D41, Q13 Vgs).



MAIN BATTERY CONNECTOR



Rail-hold: charger power (dock or barrel jack)
turns the MCU rail on and keeps it on.
DC-IN-20V to GND ~130k (R175+R194 || R199+R200)
= dock presence signature.

R106 sets charging current
100 mOhm 1.2 A

Sheet: /BATTERY-CHARGER/	
File: Battery-Charger.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 33/16